

Recent Advances in Our Understanding of Rhodopsin and Phototransduction

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Abstract—The present models of phototransduction for vertebrates and invertebrates have been reviewed and the relative literature updated. The emerging picture for vertebrate phototransduction is a result of a better knowledge of its general outlines, although some important details such as the role of calcium ions are still lacking. The molecular events involved in the rising phase of the electrical response have basically been understood, whilst those involved in response inactivation and recovery remain to be elucidated. In an overall strategy, the phototransduction in invertebrates shares a great deal of similarity with that in vertebrates but differs in the underlying molecular events. However, a complete picture of phototransduction in invertebrate photoreceptors has not yet emerged.

The available data on the structure of the visual pigment rhodopsin reveal further details on the present model of the retinal-binding pocket of the protein and consequently of the “red shift” of the absorbance of retinal.

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The problem of the energy supplied during photoreception, in particular, the availability of ATP in the rod outer segment and the presence in the disk membranes of a Ca-ATPase are discussed. Finally, recent progress in understanding the molecular mechanisms of inherited retinal diseases and relative gene identification are summarized. © 2001 Elsevier Science Ltd. All rights reserved

1. INTRODUCTION

The absorption of light by the visual pigment rhodopsin starts the molecular events of photo-transduction which terminate by inducing an electrical signal in the plasma membrane of photoreceptor cell.

The photoreceptors of the vertebrate retina are rods and cones, respectively, associated with scotopic (nocturnal) and photopic (diurnal) vision. Rods are more sensitive to light than cones, while cones have more rapid photoresponses and light adaptation (Yau, 1994; Baylor, 1996). These

differences in photoresponse patterns between rod and cones should originate from the different properties of visual pigments whose function is similar, but whose aminoacid sequences are different from each other.

The main features of the electric response to light and related biochemical cascade have been obtained from rods. Figure 1(a) shows the characteristically long and cylindrical rod cell with the main elements: the outer segment, containing stacks of flattened saccules of membranous disks and the inner segment, containing the nucleus, mitochondria and all the mechanisms necessary

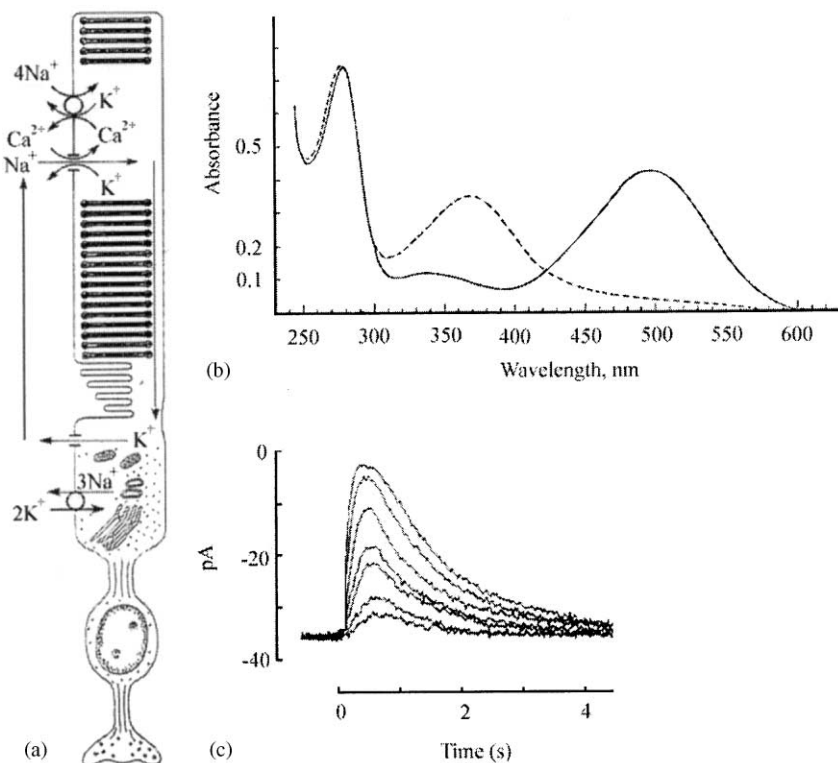


Fig. 1. (a) Schematic drawing of a rod cell of the vertebrate retina with the diagram of circulating dark current. (b) Absorbance spectra of bovine rhodopsin, before (solid line) and after light absorption (dashed line) (modified from Shichi *et al.*, 1969). (c) Electrical response of rod outer segment of the salamander retina to flashes of light of increasing intensity (modified from Baylor and Nunn, 1986).

for the cell maintenance. The cytoplasmic continuity between the inner and the outer segment occurs only through a short cilium.

In the dark, a light-sensitive current, carried by Na^+ and Ca^{2+} , enters the outer segment membrane and flows along the cell before leaving through the inner segment membrane carried mainly by K^+ . Ionic gradients of Na^+ and K^+ are maintained by the activity of an ATP-driven $3\text{Na}^+/2\text{K}^+$ pump located in the inner segment, whereas Ca^{2+} is extruded by a $4\text{Na}^+/1\text{Ca}^{2+}$, 1K^+ exchanger which depends on the energy supplied by transmembrane Na^+ gradient.

The molecular events of phototransduction take place in the rod outer segment (ROS). The disk membranes contain a very high concentration of the visual pigment rhodopsin, which absorbs light maximally at about 500 nm, as shown in Fig. 1(b). Rhodopsin is made up of a specific protein, opsin, bound to a molecule of 11-*cis* retinal. Upon the absorption of a photon, 11-*cis* retinal is first isomerized to all-*trans* retinal and then detached from the opsin, causing a decrease in the absorbance maximum (bleaching). A few milliseconds after the retinal isomerization, the activated rhodopsin molecule triggers the cyclic GMP cascade of phototransduction which ends with a signal of electrical hyperpolarization in the plasma membrane of the rod. The intensity-response relation of the light-sensitive current to light flashes of increasing intensity is shown in Fig. 1(c). Actually, light suppresses the inward current flowing in the dark across the outer segment membrane. As a consequence, the membrane potential, which in darkness is about -40 mV , hyperpolarizes, reaching a maximum of about -60 mV with saturating flash intensity (Hagins *et al.*, 1970; Penn and Hagins, 1972).

The quantitative analysis of the electrical response to a single photon allows the calculation of the energy amplification of the phototransduction process. In a toad rod, the single-photon response has a maximum amplitude of about 1 pA (Rieke and Baylor, 1998), that is one photon suppresses about 6×10^6 Na ions per second. The integration of the response area gives a charge of about 1 pC and, taking into account the value of 40 mV for the membrane potential, the corresponding electrical work can be calculated as

$(0.03\text{ V} \times 10^{-12}\text{ C} / 1.6 \times 10^{-19}\text{ C/e}^-) = 2.5 \times 10^5\text{ eV}$. While the energy carried by a single photon of wavelength 500 nm is about 2.5 eV, the energy amplification given by the cell is of the order of 100,000.

2. RHODOPSIN

Rhodopsins are made up of specific proteins, opsins, bound to retinal via a Schiff-base linkage with the ϵ -amino group of a lysine residue of the apoprotein. The variability and multiplicity of visual pigments are due to a plurality of opsins as well as the two types of retinal, namely retinal and dehydroretinal. The current nomenclature refers to rhodopsin as the visual pigment based on retinal as prosthetic group or to porphyropsin when based on dehydroretinal.

Cattle rhodopsin, being more easily obtained, is the most investigated visual pigment. In order to obtain purified rhodopsin, rod outer segments are isolated from the other cells of the retina by sucrose gradient centrifugation (Schnetkamp and Daemen, 1982). Intact disks containing rhodopsin are obtained by osmotically shocking rod outer segments, followed by flotation on 5% Ficoll (Smith *et al.*, 1975). Rhodopsin is brought into solution from the disk membranes by using a mild detergent (about 2% digitonin or cetyltrimethylammonium bromide or octylglucoside). Further purification and delipidation are performed by using affinity chromatography on a concanavalin A column (Litman, 1982). For detailed studies on the complex structure-function relationship, there is a need for functional expression of recombinant protein. Recombinant opsin has been produced by transfection of COS-1 and human embryonic kidney 293 cells (Reeves *et al.*, 1996) by injection of *Xenopus laevis* oocytes (Khorana *et al.*, 1988), in baculovirus-infected Sf9 insect cells (Janssen *et al.*, 1988), by RNA translation in wheat germ extracts (Zozulya *et al.*, 1990), and in *Saccharomyces cerevisiae* (Mollaghababa *et al.*, 1996). Recently, a method was described for the functional expression of bovine opsin in *Pichia pastoris*, a methylotrophic yeast capable of producing considerable quantities of foreign protein (Abdulaev and Ridge, 2000).

Rhodopsin is a single polypeptide of molecular weight 39 kDa, consisting of 348 aminoacids and containing two oligosaccharide chains (Hargrave, 1977; Hargrave *et al.*, 1983). It crosses the membrane with seven α -helices which constitute as much as 60% of its secondary structure and which appear oriented mostly perpendicular to the plane of the disk membrane (Unger and Schertler, 1995; Unger *et al.*, 1997).

The rhodopsin chromophore, 11-*cis* retinal, is located in a hydrophobic pocket between the helices and covalently attached to lysine 296 in helix VII via a protonated Schiff base linkage (Bownds, 1967; Wang *et al.*, 1980; Findlay *et al.*, 1981; Thomas and Stryer, 1982). The covalent bond of the chromophore contributes to the tightly held rhodopsin in a nonsignaling conformation. The extracellular domain of rhodopsin is relatively rigid, which may help to reduce spontaneous activation of the receptor in the absence of light (Schertler, 1998).

The primary step of vision, upon the absorption of light, is the isomerization of 11-*cis* retinal to all-*trans* retinal, which seems to be accompanied by a proton transfer (Becker, 1988). After some minutes, retinal is detached from the apoprotein opsin by the hydrolysis of the Schiff base, caused by a broad conformational change of the protein and the consequent access of the aqueous medium to the binding pocket.

The photolysis of rhodopsin involves thermal/dark steps characterized by intermediates which were identified at very low temperatures by their absorption spectra. The first product generated by the retinal photoisomerization from 11-*cis* to all-*trans*, was believed to be bathorhodopsin, but the formation of a new first intermediate, photorhodopsin, with absorbance maximum at 558 nm, was observed in the picosecond time regime (Schichida *et al.*, 1984). Figure 2 shows the intermediates of rhodopsin photobleaching, their absorbance maxima and their decay constants, prior to the hydrolysis of the Schiff base linkage which detaches the chromophore from opsin. A key step in the phototransduction process is the formation of metarhodopsin II which is able to bind and activate the G-binding protein transducin. Metarhodopsin II appears after about some millisecond after the absorption of light, which is the right

order of magnitude of the delay of appearance of the electric signal on the plasma membrane of the rod outer segment.

2.1. Retinal-binding site and red shift

The 11-*cis* retinal chromophore of rhodopsin is centrally located in the hydrophobic core of the seven helices closer to the cytoplasmic side of the disc membrane (Abdulaev *et al.*, 1986). Its orientation is approximately parallel to the disc membrane with a small angle of 16–20° between the chromophore and the membrane plane (Harosi and Malerba, 1975). Spectroscopic methods such as resonance Raman, Fourier transform infrared and nuclear magnetic resonance (Birge, 1990a; Siebert, 1990) or techniques of rhodopsin reconstitution with retinal analogues (Nakanishi and Crouch, 1995) and photochemical crosslinking (Nakayama and Korana, 1990; Zhang *et al.*, 1994) have been widely used to investigate the retinal-binding domain of bovine rhodopsin.

One of the problems with the absorbance spectra of rhodopsins that remains to be elucidated is the “red shift” of the absorbance of the retinal. In fact, the absorbance maximum of 11-*cis* retinal, which is about 380 nm free in solution, when it binds to opsins is displaced to maximal absorbances ranging from 435 nm of frog rods to 560 nm of human cones. At present, it seems well established that the Schiff base linkage between 11-*cis* retinal and opsin is protonated (Hubbard, 1969; Shichi, 1973; Oseroff and Callender, 1974; Barry and Mathies, 1987; Lugtenburg *et al.*, 1988) and this accounts for an absorbance shift of about 70 nm to the red. In fact, it is well known that the retinal combines with many compounds containing amino groups to give Schiff bases which absorb at about 440 nm in acid solutions (Collins, 1953; Morton and Pitt, 1955). In order to obtain a further absorbance shift (opsin shift) to about 500 nm as for bovine rhodopsin, the environmental effects of the protein on the protonated Schiff base have been considered (Kropf and Hubbard, 1958; Suzuki and Kito, 1972; Kakitani *et al.*, 1985). The lysine 296 to which retinal is bound in the seventh helix is predicted to be buried in an hydrophobic pocket near the center of the disk membrane (Hargrave *et al.*, 1983). Bringing

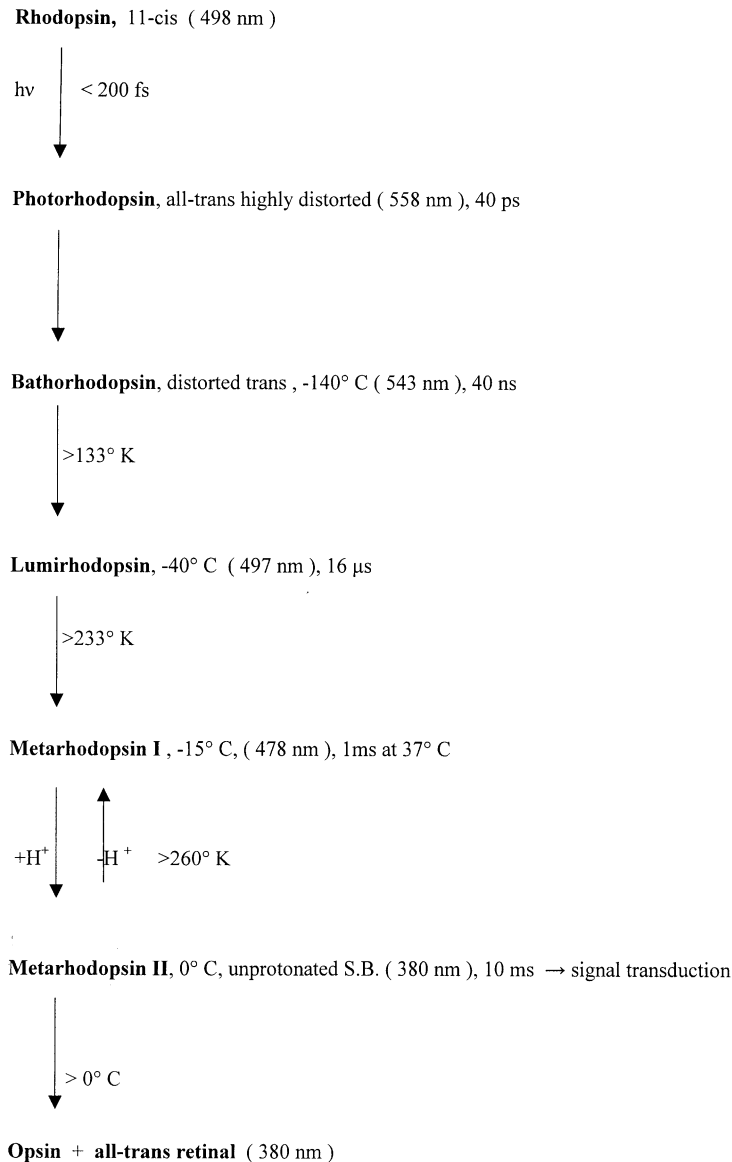


Fig. 2. Photobleaching intermediates of vertebrate rhodopsin. Absorbance maxima are given in parenthesis. Transition temperatures from one intermediate to the next one are shown by the arrows. Decay constants at room temperature are indicated for each intermediate.

negatively charged groups of some amino acid residue near the Schiff base could induce a greater degree of delocalization of the π electron system of retinal and consequently, give rise to a red shift in absorbance. Glu 113 was found to be the counter ion to the protonated Schiff base (Zhukovsky and Oprian, 1989; Sakmar *et al.*, 1989; Nathans, 1990). In fact, when Glu 113 was changed to Gln by site

directed mutagenesis, a dramatic shift in the absorbance maximum of bovine rhodopsin from 500 to 380 nm was observed. Recent resonance Raman studies on chicken cones concluded that the opsin shift between the green and red cone visual pigments arises both from electrostatic stabilization of the Schiff base-counter ion complex and from dipolar interactions near the ionone

rings (Lin *et al.*, 1994). The long-range electrostatic interactions of dipolar residues are capable of inducing absorbance shifts to either longer or shorter wavelengths depending on the residue location in the vicinity of the ionone ring or the Schiff base, respectively. Mutagenesis of rhodopsin indicates that dipolar effects from substitutions at nine positions simultaneously (M86L, G90S, A117G, E122L, A124T, W265Y, A292S, A295S, and A299C) could account for the opsin shift from 500 to 438 nm (Lin *et al.*, 1998).

In the present model of the retinal-binding pocket of bovine rhodopsin (see Fig. 3), the carboxylic acid group of Glu 113 on helix III is positioned near the Schiff-base nitrogen and interacts with the 12th carbon atom of the retinal hydrocarbon chain (Birge *et al.*, 1988; Birge, 1990b; Han *et al.*, 1993). Key residues that interact with retinal also include Gly121 on helix III and Phe261 on helix VI, while the β -ionone ring of retinal is oriented towards the residue Trp265 on helix VI (Han *et al.*, 1996). The intraligand distances of 11-*cis* retinal in the hydrophobic

binding pocket were recently measured by using magic angle spinning NMR. The C10–C20 and C11–C20 distances resulted in 0.30 and 0.29 nm, respectively, indicating a tight binding pocket, well defined to bind specifically only one enantiomer out of four possibilities (Verdegem *et al.*, 1999). The binding cavity is closed towards the intracellular side by the long and highly tilted helix III, and towards the extracellular side by the loop linking helices IV and V. The retinal binding site is slightly closer to the extracellular side of the molecule (Schertler, 1998).

2.2. Invertebrate rhodopsin

In invertebrate retina, the visual pigment is still made up of retinal bound to an apoprotein opsin, but in this case, the absorption of light does not result in the bleaching of rhodopsin, but leads to the formation of a stable metarhodopsin. The photochemistry of rhodopsin appears to be very similar to that of the vertebrates. The conversion of rhodopsin to metarhodopsin again proceeds through a number of intermediate states. The well characterized rhodopsin extracted from the squid *Loligo*, which absorbs maximally at 493 nm, is converted by light into metarhodopsin, which in the alkaline form absorbs maximally at 380 nm and in the acid form at about 500 nm (Hubbard and St. George, 1958). As for the vertebrates, squid rhodopsin contains 11-*cis* retinal as chromophore, while the photoproduct metarhodopsin contains the all-*trans* isomer.

After *Loligo*, other cephalopods rhodopsins were extracted and characterized such as those from *Todarodes pacificus* (Hara and Hara, 1967), *Octopus vulgaris* (Brown and Brown, 1958) and *Eledone moschata* (Schwemer, 1969). The common finding for all these visual pigments was that the light-induced reactions stop at the metarhodopsin intermediate, with the retinal remaining attached to the opsin. Resonance Raman studies on octopus rhodopsin lead to the conclusion that this rhodopsin differs from bovine rhodopsin in the specific nature of the retinal–protein interaction (Deng *et al.*, 1991).

Similar characteristics were also found for the rhodopsins in insects (Goldsmith, 1972; Hamdorf, 1979). In *Drosophila*, the predicted existence of five

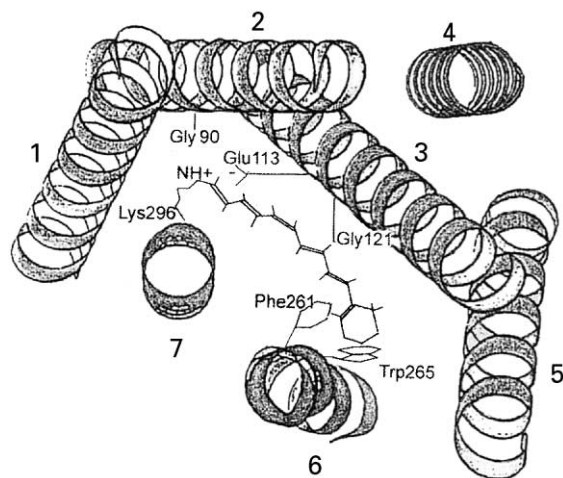


Fig. 3. The arrangement of the retinal-binding pocket between the alpha helices in bovine rhodopsin viewed from the cytoplasmic site. The seven transmembrane helices are supposed to have a clockwise sequential order. At the center of the figure, the chromophore 11-*cis* retinal is bound to Lys296 on the helix VII via a protonated Schiff base linkage, which is drawn near the counter ion Glu113. Residues that interact with retinals such as Gly 90 on helix II, Gly121 on helix III, Phe261 and Trp265 on helix VI are also shown (modified from Schertler, 1998).

rhodopsins expressed in the separated populations of photoreceptors have now been realized through the molecular identification of a family of five opsin genes expressed in the compound eye (see review article of Montell, 1999). The major rhodopsin, which is expressed in the R1-6 cells, absorbs maximally at 480 nm. The blue- and green-sensitive photopigment Rh5 and Rh6 absorb maximally at about 440 and 520 nm (Montell, 1999) and the two ultraviolet pigments Rh3 and Rh4 have absorbance maxima at 345 and 375 nm, respectively (Feiler *et al.*, 1992). Many other insects' eyes are able to respond to ultraviolet light. The rhodopsin extracted from the retina of *Ascalaphus macaronius* showed an absorbance maximum at 345 nm, about 30 nm shorter than the λ_{max} of the free retinal (Hamdorf *et al.*, 1971). The stereoconfiguration of retinal in this ultraviolet pigment was found to be 11-*cis* and the Schiff-base linkage was supposed to be unprotonated to explain the hypsochromic shift found for the absorbance maximum (Paulsen and Schwemer, 1972). Surprisingly, resonance Raman studies on rhodopsin₃₄₅ from blowfly lead to the conclusion that the Schiff-base linkage between 11-*cis* retinal and opsin is instead protonated (Pande *et al.*, 1987).

Investigations on the molecular structure of invertebrate rhodopsin have been developed for insects such as *Drosophila* (Zuker *et al.*, 1985) and *Calliphora* (Huber *et al.*, 1990), and cephalopods such as *octopus* (Ovchinnikov *et al.*, 1988) and *Loligo* (Hall *et al.*, 1991; Davies *et al.*, 1996). Pictures similar to bovine rhodopsin were obtained. The invertebrate opsin is a single polypeptide with seven alpha-helices which span the cellular membrane. Its length is somehow different from all rod rhodopsins which have the same number of 348 amino acids: rhodopsin in *Drosophila* is made up of 373 amino acids, while in *octopus* it is about 455.

2.3. Rhodopsin regeneration

The absorption of light by rhodopsin leads to the *cis*-to-*trans* isomerization of the 11-*cis* retinal chromophore and, in the vertebrate eye, the hydrolysis of the Schiff-base linkage between retinal and opsin. As the chromophore retinal is

released in all-*trans* form, the process of regeneration, which takes place in the dark, requires the all-*trans* retinal to be isomerized to the 11-*cis* form, which will spontaneously recombine with opsin to give rhodopsin (Hubbard, 1956; Amer and Akhtar, 1973; Rando *et al.*, 1991). The released all-*trans* retinal is rapidly converted in the outer segment by a specific dehydrogenase to all-*trans* retinol (Rattner *et al.*, 2000), which is delivered *via* binding proteins to the pigment epithelium. Within the pigment epithelium, all-*trans* retinol is isomerized to 11-*cis* retinol by a retinol isomerase (Bernstein *et al.*, 1987; Bridges, and Alvarez, 1987; Deigner *et al.*, 1989; Stecher *et al.*, 1999). An alcohol dehydrogenase is then required in order to produce 11-*cis* retinal which diffuses back to the ROS for rhodopsin regeneration (Futterman and Saslaw, 1961; Zimmerman *et al.*, 1975; Nicotra and Livrea, 1982; Simon *et al.*, 1995).

Another process which restores the visual pigment content is the renewal of the photoreceptor membranes in which both the protein and lipid are being continually replaced. A newly synthesized molecule of opsin binds to 11-*cis* retinal forming rhodopsin, which is incorporated into the new disk membrane. These disks are pushed up the outer segment by the renewal process. After 5–80 days, depending on the species, the disks disappear from the top of rod outer segment and are phagocytosed by the pigment epithelial cells (Bok and Young, 1972; Young, 1974; Anderson *et al.*, 1980; Andrews, 1984; Roofs, 1986).

The net synthesis of rhodopsin seems light-intensity-dependent and can adjust when an animal encounters a new lighting environment. The number of molecules added per disk increases to ensure the appropriate packing density of rhodopsin for the increased level of light intensity (Schremser and Williams, 1995).

It seems that the rod outer segment renewal in rods is mediated by polarized sorting of rhodopsin on post-Golgi vesicles that bud from the *trans*-Golgi network and fuse with the specialized domain of the plasma membrane in the rod inner segment. This domain surrounds the cilium that connects the inner segment and the rod outer segment to which mature rhodopsin is delivered. Small GTP-binding protein rab6, as well as the carboxyl-terminal domain of rhodopsin

are essential in the formation of post-Golgi vesicles (Deretic, 1998).

In invertebrate visual systems, rhodopsin and metarhodopsin are photoconvertible. All-*trans* retinal, the chromophore of metarhodopsin is converted by light to the 11-*cis* isomer (Hubbard and St. George, 1958; Brown and Brown, 1958). Therefore, rhodopsin can be obtained again by irradiating metarhodopsin with wavelengths in the range of its absorption. This fast process, called photoregeneration, plays an essential role in maintaining the rhodopsin content, and thus the receptor sensitivity, at a high level (Hamdorf and Schwemer, 1975; Hillman *et al.*, 1983).

The visual pigments of invertebrates are also regenerated by a slow process of membrane renewal quite similar to that which occurs in vertebrates (White, 1968; Schwemer, 1986). In this process, the presence of 11-*cis* retinal is crucial for the biosynthesis of rhodopsin (Paulsen and Schwemer, 1983).

In the well studied compound eye of the blowfly, continuous illumination leads to a photoequilibrium between the rhodopsin and metarhodopsin; the ratio between the concentration of rhodopsin and that of metarhodopsin depends on the spectral composition of the environmental light and on the extinction coefficients of the two pigments. The light intensity affects only the kinetics of the photoreactions (Hamdorf and Schwemer, 1975; Hillman *et al.*, 1983). However, the alternation of the two thermostable states of rhodopsin and metarhodopsin under irradiation, is not of unlimited duration because of the continuous renewal which implies degradation and resynthesis of molecules. Rhodopsin degrades with a mean lifetime of about 5 days, while metarhodopsin decays 60-fold more rapidly (Schwemer, 1984). Experiments on the blowfly ommatidia clearly demonstrated that retinal isomerization is a light-dependent process with maximal effect when the animals are kept in the light of the blue-violet range (about 440 nm). This indicates that the all-*trans* retinal released in the cell by the metarhodopsin degradation binds to a protein, reaching in this way the two-fold effect of shifting its absorbance maximum from 370 nm to the visible range and making a highly stereospecific retinal photoisomerization possible.

Two well characterized examples of this kind of retinal-binding proteins are the retinochrome of cephalopods (Hara and Hara, 1965, 1973; Seki *et al.*, 1982; Tokunaga and Yoshihara, 1995) and the retinal photoisomerase of honeybees (Pepe and Cugnoli, 1980; Schwemer *et al.*, 1984; Pepe *et al.*, 1982; Cugnoli *et al.*, 1989; Smith and Goldsmith, 1991; Pepe *et al.*, 1990). These proteins are able to direct the photoisomerization of all-*trans* retinal almost exclusively toward the 11-*cis* retinal formation. Moreover, when all-*trans* retinal as the starting isomer was replaced by 13-*cis* or 9-*cis* retinal, the favored product of the irradiation was still 11-*cis* retinal (Hara and Hara, 1973; Cugnoli *et al.*, 1989).

3. VERTEBRATE PHOTOTRANSDUCTION

In the present model of vertebrate phototransduction, a dark current carried by Na^+ and Ca^{2+} enters the outer segment of the rod cell through cGMP-gated channels keeping the cytoplasmic membrane relatively depolarized. Light activates a specific phosphodiesterase which hydrolyses cGMP to 5'-GMP causing the ionic channels to close and the dark current to be interrupted. In this way, the absorption of light is transformed by the rod cell into a hyperpolarized electrical signal, which is transmitted to the synaptic terminal where it slows the transmitter release. The so-called cGMP cascade underlying phototransduction is known in detail (see Fig. 4). Light is absorbed by the 11-*cis* retinal chromophore of rhodopsin causing the apoprotein to become catalytically active. The activation probability of rhodopsin is about 0.7 after a single photon encounter (Birge, 1990a). The photoexcited state (R^*), which corresponds to the spectroscopic intermediate metarhodopsin II, is able to activate the GTP-binding protein transducin (T), which in turn activates a phosphodiesterase (PDE) specific for cGMP.

The speed of the interaction between R^* and transducin is explained by the high mobility of rhodopsin molecule in the plane of the disk membrane. Within a fraction of a second, one molecule of R^* is able to contact hundreds of transducins and cause the exchange of bound

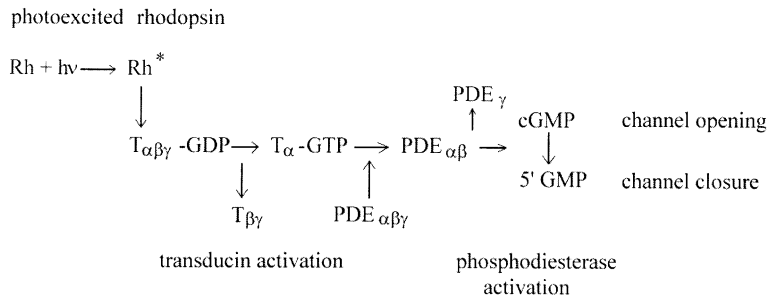


Fig. 4. Schematic representation of the vertebrate phototransduction cascade.

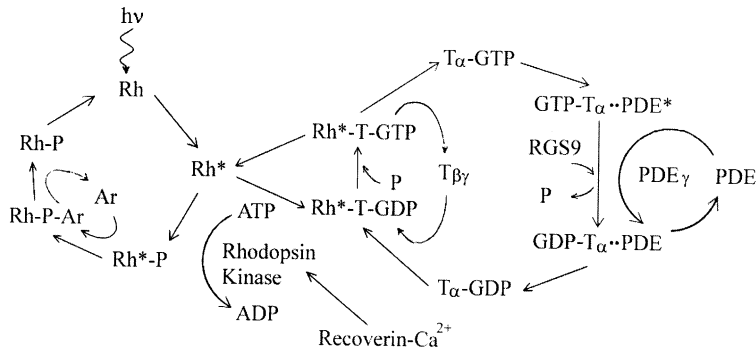


Fig. 5. Detailed scheme of vertebrate phototransduction process. On the left, rhodopsin cycle is shown with an activation by light and inactivation through phosphorylation and arrestin binding (Ar). In the middle, transducin activated by Rh^* exchanges GDP for GTP and is inactivated by its own GTPase activity accelerated by RGS9 or PDE_{γ} . On the right, PDE is activated by $T_{\alpha}\text{-GTP}$ and loses its γ subunit.

GDP for GTP on its α subunit T_{α} , providing the first stage of the cascade amplification. One molecule of the complex $GTP\text{-}T_{\alpha}$ is able to activate one molecule of the PDE by removing its inhibitory γ subunits. The second stage of amplification of the phototransduction cascade is provided by the activity of PDE which catalyzes the hydrolysis of about 1000 molecules of cGMP per second (Miki *et al.*, 1973; Lamb, 1996). The total amplification of the cGMP cascade results to be about 10^2 (first stage) $\times 10^3$ (second stage) $= 10^5$ in agreement with the calculations based on the electrical response.

A more detailed scheme of phototransduction is shown in Fig. 5. Activated rhodopsin (Rh^*) binds to T_{α} and catalyzes the exchange of GDP for GTP. $T_{\alpha}\text{-GTP}$ in turn activates PDE by removing the inhibitory subunit PDE_{γ} . The falling phase of the electrical response needs the inactivation of rhodopsin through the phosphorylation by a

kinase and the subsequent binding of arrestin (Ar). Transducin also gives its contribution to the switch-off mechanism through its GTPase activity which is accelerated by the coordinated action of the G-protein RGS9 and the γ subunit of PDE.

3.1. Rhodopsin activation

Recent studies have given further details on the central role of rhodopsin in the phototransduction cascade. In the dark, rhodopsin is constrained in an inactive conformation by the interactions with 11-*cis* retinal. Upon photoisomerization, major structural rearrangements lead to the formation of the active form of the protein, metarhodopsin II (Kibelbek *et al.*, 1991). The light-activated conformational changes seem to involve the rigid body motion of helix VI relative to helix III, rather than secondary structural changes (Farrens *et al.*, 1996), while an inward motion of helix VII toward

the cytoplasm also occurs (Farrens *et al.*, 1996; Abdulaev and Ridge, 1998). As a consequence of this motion, changes would be expected in the third cytoplasmic loop recognized by transducin and rhodopsin kinase. A mechanism for how a retinal isomerization leads to rhodopsin activation seems to involve a direct steric interaction between retinal chromophore and transmembrane helix 3 in the region of Gly121 (Shiekh *et al.*, 1997) and rotations of Glu134, Tyr223, Trp265, Lys296 and Tyr306 and rearrangements of their H bonds (Pogozheva *et al.*, 1997). Glu 134 is a highly conserved residue and is an important link in the appearance of the transducin binding domain in the cytoplasmic loops of rhodopsin (Fahmy *et al.*, 1995). Moreover, studies made on interactions of opsin and 10-methyl homologue of 11-*cis* retinal, provide a direct support for the concept that the relaxation of strain in the retinal polyene chain acts as the major driving force of the cascade dark reactions (DeLange *et al.*, 1998). The salt bridge between Lys296 and Glu113 helps to keep opsin in the inactive state. The breaking of this bridge, following deprotonation of the Schiff base in metarhodopsin II, activates the protein (Cohen *et al.*, 1992; Shiekh *et al.*, 1997). An unexpected result was found when the active site Lys296 was changed to Gly. This mutant was able to activate transducin in the absence of the retinal chromophore (constitutive activity) (Robinson *et al.*, 1992). Furthermore, activation may be obtained without illumination by the addition of all-*trans* retinal or its analogues to opsin provided one or more cytoplasmic carbonyl groups of opsin were protonated (Buczylko *et al.*, 1996). More evidence has accumulated to confirm that opsin or some of its mutants, when combined with all-*trans* retinal, can activate transducin (Jager *et al.*, 1996; Surya and Knox, 1998; Han *et al.*, 1998).

3.2. Transducin

Transducin is a membrane-bound protein quite similar structurally and functionally to other heterotrimeric G proteins and is made up of three subunits T_α , T_β , T_γ . Its activation involves the binding to amino acids located in the second and the third cytoplasmic loops of metarhodopsin (Franke *et al.*, 1992; Konig *et al.*, 1989; Ernst

et al., 1995; Shi *et al.*, 1995). It seems in fact that the α -subunit of transducin could be cross-linked to metarhodopsin II to position 240 (Resek *et al.*, 1994). Due to the recent resolution of crystal structures of T_α , the interaction between rhodopsin and T_α have been intensively studied leading to a main model system (Hamm *et al.*, 1988; Martin *et al.*, 1996; Onrust *et al.*, 1997). Metarhodopsin II catalyzes a conformational change in T_α with the opening of the nucleotide-binding pocket and release of GDP. This state of "empty pocket" leads to a high affinity binding between metarhodopsin II and T_α and the site of this interaction seems located in the carboxyl terminus of T_α (Hamm, 1998). The site that directly provides the signal from rhodopsin to activate GDP release from T_α has been proposed to be at the highly conserved residues Glu134 and Arg135 (Acharya and Karnik, 1996).

Also the conserved disulfide bond between Cys110 and Cys187 connecting helix III to the second intradiscal loop seems involved in determining the stability of metarhodopsin II and its coupling to transducin activation (Davidson *et al.*, 1994). More recently, the role of the four cytoplasmic loop of rhodopsin has been established. It seems that the amino-terminal region of this loop is a part of the binding site for the carboxyl terminus of the α subunit of transducin and plays a role in the regulation of $\beta\gamma$ subunit binding (Ernst *et al.*, 2000).

When GTP replaces GDP, a further conformational change of T_α causes a decrease of the affinity for rhodopsin. T_α -GTP dissociates from R^* and activates PDE on disk membranes on a timescale of a few milliseconds.

3.3. Phosphodiesterase

CGMP-specific phosphodiesterase (PDE) is a membrane-bound protein consisting of four subunits, two catalytic and two inhibitory. In the dark-adapted rods, the catalytic $P_{\alpha\beta}$ subunits of PDE are blocked by two identical inhibitory subunits P_γ . In the light, PDE is activated by T_α -GTP which binds to P_γ subunits and displaces them from the catalytic $P_{\alpha\beta}$ (Stryer, 1996; Granovsky *et al.*, 1997). A combination of a photo-cross-linking approach with fluorescence assays in

order to probe the $P_\gamma/P_{\alpha\beta}$ interface has yielded new information about the inhibitory site on PDE and the mechanism of the inhibition by P_γ . The site that binds the P_γ C terminus is within the PDE catalytic domain, where cGMP binds and corresponds to residues 751–763 of P_α (Artemyev *et al.*, 1996).

PDE catalyzes the hydrolysis of cGMP into 5'GMP. The rate has been measured on purified PDE (Cook and Beavo, 2000). Values of turnover range from about 600 to 4200 cGMP/molecule PDE/s (Pugh and Lamb, 1993), although it is unlikely that this rate is reached in the photoreceptor cell since the average concentration of free cGMP is about 10 μ M.

In addition to the catalytic site where cGMP is hydrolyzed, the $P_{\alpha\beta}$ subunits of PDE also contain high-affinity noncatalytic binding sites (Yamazaki *et al.*, 1980). The role of these noncatalytic site is not clear for photoreceptor PDE. However, the occupancy of the noncatalytic sites with cGMP seems to increase the binding affinity of P_γ for $P_{\alpha\beta}$ (Arshavsky *et al.*, 1992; Yamazaki *et al.*, 1996). In the dark, much of the total cGMP content in the rod outer segment is effectively sequestered by the high-affinity noncatalytic sites. In the light, on activation of PDE by transducin, these sites undergo changes in binding affinity and dissociation kinetics (Cote *et al.*, 1994), but the significance of these changes in terms of regulatory actions on the catalytic site of the enzyme is not yet clear.

3.4. Guanylate cyclase

The steps described above are involved in the rising phase of the electrical response and are basically understood, while those involved in response inactivation and recovery instead remain to be elucidated. After a flash of light, the cytoplasmic concentration of calcium ions decreases approximately from 0.5 to 0.1 μ M (Gray-Keller and Detwiler, 1994; Schnetkamp, 1995) due to the fact that Ca^{2+} is excluded from the rod outer segment by the $Na^+/Ca^{2+},K^+$ exchanger, while it cannot enter through the closed channels. The decrease of the calcium level stimulates the activity of a guanylate cyclase which synthesizes cGMP for the reopening of the cationic channels which in turn causes the falling phase of the

electrical signal. Therefore, the role of guanylate cyclase in photoreceptor is to maintain sufficiently free cGMP in the cytoplasm so that an adequate number of cGMP-activated cation channels are open and the dark current is restored (see Fig. 6).

The guanylate cyclase of photoreceptor ROS belongs to the family of particulate guanylate cyclase and is an oligomeric enzyme comprising of a 115 kDa integral membrane protein (Hayashi and Yamazaki, 1991; Johnston *et al.*, 2000). The activity of the enzyme significantly increases 5–20 times when calcium ion concentration decreases from about 500 nM to below 100 nM (Lolley and Racz, 1982; Pepe *et al.*, 1986; Koch and Stryer, 1988; Koutalos *et al.*, 1995a). The Ca^{2+} -dependent regulation is mediated by Ca^{2+} -binding proteins GCAP1 and GCAP2 (Gorczyca *et al.*, 1994), which belong to a subfamily of neuronal Ca^{2+} -binding proteins that originate from a common calmodulin-like ancestor (Polans *et al.*, 1996). The Ca^{2+} -binding protein S100, which was also found to be present in ROS of bovine photoreceptors, seems able to activate guanylate cyclase at μ M calcium concentration, suggesting for S100 protein an important role in the cGMP synthesis of dark-adapted photoreceptors (Poznyakov *et al.*, 1995; Rambotti *et al.*, 1999). In

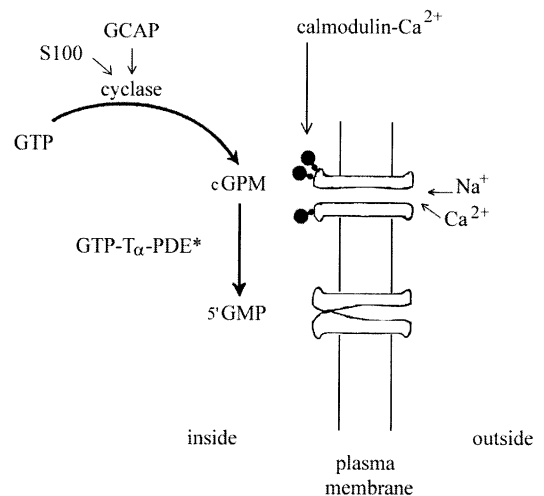


Fig. 6. Schematic drawing of the opening and closure of cGMP-gated channels. The intracellular concentration of cGMP is controlled by the activities of PDE and guanylate cyclase, which in turn is regulated by Ca^{2+} -binding proteins GCAPs or S100.

summary, the activity of guanylate cyclase can be switched on by two distinct calcium signals: one in the range of 500–100 nM, *via* GCAP1, and the second signal in the range of micromolar, *via* the binding of an S100 protein. The low calcium signal is linked to phototransduction, while the high calcium signal is not yet clear (see also the excellent review on photoreceptor guanylate cyclase of Pugh *et al.*, 1997). To complicate the picture, a newly discovered protein in frog retina, which shares conserved sequences with the GCAPs, inhibits the cyclase when the Ca^{2+} increases (Li *et al.*, 1998).

3.5. Rhodopsin phosphorylation

At the same time, when cGMP levels are restored by the cyclase activity, shutoff reactions give their contribution to the recovery of dark conditions: R^* is stopped by phosphorylation followed by rhodopsin binding to the soluble protein arrestin (Palczewski *et al.*, 1992); active T_α is shut-off by its own GTPase activity, while PDE is inactive when it recombines with its inhibitory γ subunit.

Considerable information is now available on rhodopsin phosphorylation. Photoactivated rhodopsin (R^*) is phosphorylated near the C terminus at multiple sites, predominantly at Ser334, Ser338, and Ser343 (Ohguro *et al.*, 1996). When rhodopsin becomes triphosphorylated it appears to become phosphorylated on threonine residues such as Thr335, Thr336, Thr340, and Thr342 (Ohguro *et al.*, 1993, 1996; McDowell *et al.*, 1993; Zhang *et al.*, 1997). It seems that the major role of rhodopsin phosphorylation, mediated by rhodopsin kinase (RK) and protein kinase C (Udovichenko *et al.*, 1997), is to promote high-affinity arrestin binding and decrease transducin binding. In this way, the mechanism of arrestin quenching of phototransduction is *via* steric exclusion of transducin binding to phosphorylated R^* (Krupnick *et al.*, 1997). The binding region of arrestin for phosphorylated rhodopsin is located in the N-terminal domain (Granzin *et al.*, 1998), precisely in the region composed of, at least in part, aminoacids 101–130 (Smith and Hargrave, 2000). Although rhodopsin phosphorylation at multiple sites can support arrestin binding (Zhang *et al.*,

1997), only one or two sites are thought to be necessary for arrestin binding *in vivo* (Palczewski, 1997). Rhodopsin phosphorylation is regulated by recoverin, a 23 kDa Ca-binding protein found predominantly in vertebrate photoreceptor cells. Recent studies indicate that recoverin inhibits rhodopsin phosphorylation by directly regulating rhodopsin kinase (Chen *et al.*, 1995). It was found that Ca^{2+} -bound recoverin forms a complex with rhodopsin kinase at the disk membrane surface and remarkably suppresses rhodopsin kinase activity (Sanada *et al.*, 1996). Half-maximal inhibition of rhodopsin phosphorylation by recoverin occurs at 1.5–3 μM [Ca^{2+}]. This is consistent with the current model of phototransduction in which the fall in free Ca^{2+} in ROS following illumination exerts a negative feedback by relieving inhibition of rhodopsin phosphorylation (Klenchin *et al.*, 1995).

The *in vivo* recycling of phospho-rhodopsin to rhodopsin requires the presence of a phosphatase, which has been actually isolated from rod outer segment as a type 2A protein phosphatase (King *et al.*, 1994).

3.6. Gtpase activity of transducin

Photoreceptor recovery from the light response also requires that T_α -GTP hydrolyzes its bound GTP to GDP (Lamb and Matthews, 1988; Sagoo and Lagnado, 1997). However, the rate of GTPase activity of isolated transducin is too slow, about 0.05 s^{-1} , to account for the rapid recovery rates observed *in vivo* (Angleon and Wensel, 1993). In fact, in the photoreceptor cell this rate is accelerated by a coordinated action of two proteins, PDE γ (Tsang *et al.*, 1998) and the G-protein RGS9 (Cowan *et al.*, 1998). In photoreceptors, RGS9 exist as a complex with the long splice variant of type 5 G protein β subunit $\text{G}\beta_{5\text{L}}$ (Makino *et al.*, 1999). The role of RGS9 is to provide transducin with the RGS homology domain, which seems to stabilize T_α in the right conformation for GTP hydrolysis as suggested from crystallographic studies (Sondek *et al.*, 1994). PDE γ itself does not activate transducin GTPase, but it enhances the catalytic action of RGS9. In physiologically intact photoreceptor cells, the rate

of GTPase activity of transducin exceeds 7 s^{-1} when maximally accelerated (Tsang *et al.*, 1998).

3.7. Cyclic nucleotide-gated channels

In ROS plasma membrane, ion channels through which Na^+ and Ca^{2+} enter to depolarize the rod are activated by the direct binding of cGMP. The relatively high cGMP concentration, which keeps the ion channels open in the dark, decreases in the light causing some of the channels to close, decreasing the cation influx and hyperpolarizing the rod cell. The native rod channel is thought to be a tetramer composed of homologous 63-kDa α and 240-kDa β subunits (Kaupp *et al.*, 1989, Liu *et al.*, 1996a). Each subunit contains six transmembrane domains followed by a cyclic nucleotide-binding domain in the C-terminal (Brown *et al.*, 1995). However, α subunits can form functional channels on their own, whereas β subunits alone are unable to form a functional channel (Chen *et al.*, 1994). Opening of the channel requires the binding of three or four molecules of cGMP. The probability for channel open increases as successive nucleotide binding sites are filled and each liganded state can produce channel openings to more than one conductance level (Ruiz and Karpen, 1997). A exhaustive model of the cGMP-gated channel is still lacking and should also take into account the effects of divalent cations. In fact, single channel recordings showed that divalent cations decrease the unit conductance to 0.1 pS which is unusually small for channel standards (Bodoia and Detwiler, 1985). When the size of the single channel current was increased by using low concentrations of divalent cations, the unit conductance increased to a value of 25 pS (Haynes *et al.*, 1986; Taylor and Baylor, 1995), in accordance with that of most of the ionic channels.

Ca^{2+} ions, whose intracellular concentration varies directly with the activity of the cGMP-activated channels, have among their targets the channels themselves. In fact, the cGMP sensitivity of the channels seems modulated by calcium and calmodulin (Hsu and Molday, 1993). The inhibition by Ca^{2+} -calmodulin is much more pronounced in olfactory channels than in rod channels. The $K_{1/2}$ of cGMP increases 2-fold in

rod when Ca^{2+} -calmodulin binds to the β subunit (Chen *et al.*, 1994), whereas it increases about 50-fold in olfactory channels (Balasubramanian *et al.*, 1996). The calmodulin binding sites in rod channels appear to be located in the β subunits: one, CaM1, in the N terminus, whereas the other site, CaM2, is located in the C terminus (Weitz *et al.*, 1998).

4. ROLE OF CALCIUM IONS

A high concentration of calcium, of the order of mM, is present in the rod outer segment for the most part sequestered in the disk membranes. The content of free calcium in the cytoplasm of the rod outer segment is instead very low: in the dark it was estimated to be about $0.2\text{ }\mu\text{M}$ by fluorescent indicators as fura 2 (Ratto *et al.*, 1988) or about $0.4\text{--}0.5\text{ }\mu\text{M}$ by aequorin (Lagnado *et al.*, 1992). At present, it seems well established that two Ca^{2+} transport systems found in ROS plasma membrane regulate the free calcium concentration in the cytoplasm. One is the cGMP-gated cation channel through which Ca^{2+} ions enter the outer segment of the photoreceptor cell carrying about 10–15% of the total current (Nakatani and Yau, 1988). The other one is the $\text{Na}^+/\text{Ca}^{2+}$, K^+ exchanger through which Ca^{2+} is extruded from the cell. The presence of a Ca^{2+} ATPase on ROS plasma membrane seems to be ruled out (Yau and Nakatani, 1985).

After a bright illumination which suppresses the dark current and therefore the calcium influx, extrusion of Ca^{2+} by the $\text{Na}^+/\text{Ca}^{2+}$, K^+ exchanger occurs at a rate of about $2\text{ }\mu\text{M/s}$. The lowest level to which calcium concentration can decrease inside the photoreceptor cell was calculated using the equation of Blaustein and Hodgkin to be of the order of 10^{-10} M (Cervetto *et al.*, 1989). However, measurements with fura 2 gave a Ca^{2+} drop from $0.5\text{ }\mu\text{M}$ to only about $0.1\text{--}0.05\text{ }\mu\text{M}$ after illumination (Ratto *et al.*, 1988; Gray-Keller and Detwiler, 1994). This discrepancy seems overcome by the report that in isolated bovine ROS kept in Ca^{2+} -free medium, the internal Ca^{2+} concentration is maintained at relatively high values (83 nM) by a combined inactivation of the exchanger coupled

with Ca^{2+} release from disk membranes (Schnetkamp, 1995).

The rod $\text{Na}^+/\text{Ca}^{2+}$, K^+ exchanger is made up of a high molecular weight polypeptide located in the plasma membrane of ROS as a homodimer. The model based on sequence analysis and labeling studies indicate that it contains two multispanning membrane domains separated by a large intracellular loop containing repeat segments. A large extracellular domain at the *N*-terminus contains multiple sialo-oligosaccharides (Kim *et al.*, 1998).

The main role of calcium ions in rod cells of vertebrate retina is to affect the cGMP metabolism during the phototransduction events. The decrease of the calcium level after illumination stimulates the activity of a guanylate cyclase which synthesizes cGMP for the reopening of the cationic channels. The same decrease of calcium concentration seems able to inhibit the phosphodiesterase activity by a factor of three (Koutalos *et al.*, 1995b).

However, the homeostasis of calcium in ROS still presents aspects that remain to be explained. Little is known about the ability of the ROS disks to exchange Ca^{2+} with cytoplasm and the significance of this in terms of photoreceptor functionality. Disks are known to store calcium at mM level with a mechanism which is not clearly understood. The uptake of Ca^{2+} by disks against its gradient is an ATP dependent process (Puckett *et al.*, 1985; Panfoli *et al.*, 1994a) associated with a Ca^{2+} -ATPase (Davis *et al.*, 1988; Panfoli *et al.*, 1994a,b). The release of Ca^{2+} from disks has been measured by many authors under different conditions. Cyclic GMP was found to stimulate the Ca^{2+} efflux (Cavaggioni and Sorbi 1981; Caretta and Cavaggioni, 1983; Caretta, 1985; Koch and Kaupp, 1985) although it has been suggested that cGMP-gated channels are exclusively located in the ROS plasma membrane (Bauer, 1988) while the experiments so far referred was run on disrupted ROS membranes containing disks as well as cytoplasmic membranes. However, a cGMP-elicited efflux of Ca^{2+} was also found in purified disks separated from cytoplasmic membranes by flotation on 5% Ficoll solution (Puckett and Goldin, 1986). More recently, (Schnetkamp, 1995) measured an increase in cytosolic calcium

concentration from about 150 nM to about 600 nM in intact ROS. As the $\text{Na}^+/\text{Ca}^{2+}$, K^+ exchanger was stopped by removing the external Na^+ , and Ca^{2+} uptake from the external medium was prevented by addition of EGTA, the increase of internal calcium level was interpreted as due to an efflux of calcium from disks.

Ca^{2+} release from disk membranes could be also due the activity of one of the isoforms of phospholipase C which was found to be present in rod outer segments (Panfoli *et al.*, 1992; Peng *et al.*, 1997), although it seems that IP_3 receptors in rods are few (Day *et al.*, 1993).

Therefore, some evidences have been accumulated indicating that the exchanges of Ca^{2+} between the disks and the cytoplasm give a contribution to the regulation of the internal concentration of Ca^{2+} in addition to the well-known mechanisms of the influx through the light-sensitive channels and the extrusion through the $\text{Na}^+/\text{Ca}^{2+}$, K^+ exchanger.

Calcium ions also play a central role in light adaptation, the mechanism by which photoreceptors exhibit flash responses that become smaller and faster in the presence of background illumination. The sensitivity to a brief flash is decreased and the kinetics of the response are accelerated. Light adaptation enables the photoreceptor to operate over an extended range of light intensities by reducing the gain of the single photon response as the light level rises (Baylor and Hodgkin, 1974). In this way, the sensitivity of a rod cell is adjusted according to the background light, enabling the photoreceptor to detect incremental stimuli over a 100,000-fold range of ambient light intensity. Results from many experiments indicate that Ca^{2+} acts as a second messenger in the regulation of photoreceptor sensitivity (see the review article of Fain *et al.*, 2001). A change in Ca^{2+} intracellular concentration is necessary for light adaptation to occur. Ca^{2+} -dependent modulation of the rate of guanylate cyclase could rescue the light-sensitive current from saturation and make it possible for the photoreceptor to respond to incremental changes in light intensity despite the PDE activity due the background light (Fain *et al.*, 2001). Calcium is also supposed to act on light adaptation by modulating the cGMP-gated channel through calmodulin interaction with the 240 K

protein, which is tightly associated with the channel (Molday *et al.*, 1990). It was reported in fact, that calmodulin binds to the 240 K protein and increases the apparent Michaelis constant of the channel for cGMP. This has the effect of increasing the sensitivity of the channel to small changes in cGMP levels during phototransduction (Hsu and Molday, 1993). However, a Ca^{2+} -dependent modulation of the cGMP-gated channels seems to play a role in light adaptation, more importantly in cones than in rods.

5. ENERGETICS OF THE PHOTORECEPTOR

The electrical response to the light of the photoreceptor cell involves an energy much greater than that transported by the absorbed photons. Early studies made on vertebrate retina reported an increase of phosphate upon illumination associated with a decrease of high energy phosphate esters (Auricchio and De Berardinis, 1952). A decrease of ATP was reported in ROS upon illumination (Robinson *et al.*, 1974). The content of high-energy phosphate esters available in the dark adapted rod outer segment of the frog retina was measured and corresponded to about 1.4 mM. This value decreased by about 60% in less than 12 s after bleaching a few percent of the rhodopsin molecules (Caretta and Cavaggioni, 1976).

The problem of availability of ATP in the rod outer segment during photoreception remains to be elucidated. It is well known that mitochondria are only present in the inner segment of the rod so that the ATP synthesized is thought to diffuse through the connecting cilium and reach the disks of the outer segment (Biernbaum and Bownds, 1985). However, the reaction times involved in the photoreceptor cell functioning are shorter than those for the diffusion of ATP molecules, which should travel a particularly difficult path through the disk membranes and reach the apical disks of the ROS where the rhodopsin cascade works (Boesze-Battaglia and Albert, 1990). In fact, the average time for a small molecule such as cGMP to diffuse through the entire outer segment was estimated to be about 6 min by fluorescent probes (Olson and Pugh 1993), while the time involved in

phototransduction reactions is of the order of milliseconds. Therefore, the synthesis of ATP is likely to happen on the ROS where the main reactions for the photoreception need an immediate source of energy. The finding that glyceraldehyde-3-phosphate dehydrogenase is present in large quantities in rod outer segments suggests that a part of the ATP and GTP synthesis in ROS takes place from glycolysis (Hsu and Molday, 1990) which produces about 95 μM ATP/s (Hsu and Molday, 1994). The need of ATP in ROS is mainly due to the synthesis of cGMP. In the dark, a value of 28 μM ATP/s was obtained for the basal cGMP turnover (Ames *et al.*, 1986). Following illumination, the cost of synthesizing cGMP increases by a factor of 4.5 (Ames *et al.*, 1986) reaching about 126 μM ATP/s, which is higher than the rate of ATP synthesized by glycolysis. Additional energy is also needed for supplying the GTP hydrolyzed by transducin and the ATP for rhodopsin phosphorylation. Therefore, some kind of shuttle which transports high-energy phosphate groups from the inner to the outer segment of the rod has been thought necessary (Hsu and Molday, 1994). Moreover, a system capable of maintaining a high local ATP/ADP ratio, based on creatine kinase and on diffusive metabolites such phosphocreatine and creatine, was shown to be present in bovine ROS (Hemmer *et al.*, 1993).

5.1. Ca-ATPase of the disks

Vanadate or lanthanum, two well-known inhibitors of ATPases with phosphorylated intermediates, inhibited the active calcium uptake, suggesting that the ATP-dependent Ca^{2+} uptake measured in disk membranes is due to the activity of a Ca^{2+} -ATPase (Panfoli *et al.*, 1994a). The presence of a Ca^{2+} -ATPase on the disk membranes was first suggested by experiments on ROS of bovine retina (Bownds *et al.*, 1971; Sack and Harris, 1977) and then localized inside the disks of toad ROS by means of a cytochemical electron microscopic method (Davis *et al.*, 1988). More recently, the Ca^{2+} pump of disk membranes was isolated as a 100 kDa phosphopeptide on polyacrylamide gel electrophoresis and classified among the SERCA (Sarcoplasmic and endoplasmic reticulum calcium ATPase) type of pumps

(Panfoli *et al.*, 1994a,b). In fact, the presence of thapsigargin, which inhibits most of SERCA pumps, had an inhibitory effect on the phosphorylated intermediate of disk Ca^{2+} ATPase. Moreover, mild proteolysis by trypsin produced two autophosphorylatable fragments of 55 and 35 kDa, typical of the trypsin-digested SERCA-pumps (Sarkadi *et al.*, 1986).

The presence of a Ca^{2+} -ATPase in ROS disks, poses interesting questions about its physiological role. The low affinity of the disk pump for Ca^{2+} (K_m of Ca^{2+} uptake is about $13\ \mu\text{M}$) is consistent with the finding of a low-affinity, high-capacity calcium buffering mechanism observed above $5\ \mu\text{M}$ [Ca^{2+}] in the outer segment cytoplasm of vertebrate retina (Lagnado *et al.*, 1992). The Ca^{2+} concentration would be kept low in photoreceptor cells not only by the action of the high-affinity low-capacity $\text{Na}^+/\text{Ca}^{2+}$, K^+ exchanger, but also by the disk Ca^{2+} -ATPase that would pump calcium ions into the disks whenever the Ca^{2+} concentration rose over the micromolar range. However, the negligible ATP-dependent Ca^{2+} -uptake between 0.1 and $1\ \mu\text{M}$ (see Fig. 3) means that, at physiological Ca^{2+} concentrations, the pump does not work to uptake calcium ions into the disks.

Although the actual reversibility of an ion-ATPase has rarely been demonstrated in eucariotic cells (Dhalla and Zhao, 1988), the Ca^{2+} -ATPase of the disks is capable of reversing its function by acting as a synthesizer of ATP at the expense of the Ca^{2+} gradient. Infact, purified ROS disks, in the presence of ADP and phosphate, released calcium and synthesized ATP at the physiological range of Ca^{2+} concentrations present in rod cytoplasm (Pepe *et al.*, 2000).

It was recently reported that ROS of bovine retina also contain an adenylate kinase activity bound to the disk membranes which is dependent on the calcium levels (Notari *et al.*, 2001).

The presence of an ATP synthase and an adenylate kinase on the disk membranes suggests that ATP is regenerated from ADP *in situ* where phototransduction events take place. Both enzymes have maximal activities of about $100\text{--}160\ \mu\text{M}$ ATP/s, at physiological Ca^{2+} concentrations between 10^{-6} and $10^{-7}\ \text{M}$. In this way, any low diffusion of metabolites and ATP molecules from ROS cytoplasm is avoided and phototrans-

ductive processes such as phosphorylation of photoactivated rhodopsin or replenishment of GTP hydrolyzed by transducin can rapidly take place on the disk surface.

6. PHOTOTRANSDUCTION AND RETINAL DISEASES

Among several inherited diseases affecting the human retina, retinitis pigmentosa, a large and heterogeneous group of retinal degenerations, has been studied intensively at the molecular level (Dryja and Li, 1995; Gregory-Evans and Bhattacharya, 1998; Rattner *et al.*, 1999). Naturally occurring mutations affect a variety of genes encoding components of the vertebrate phototransduction or the visual cycle. The number of genes involved in these molecular events have been estimated to be about $40\text{--}60$ (Frederick *et al.*, 2000). At present, gene defects producing naturally occurring animal models of retinitis pigmentosa have been identified in the genes encoding rhodopsin, α or β subunits of PDE, guanylate cyclase and RPE65, a component of the retinal pigment epithelium involved in retinoid metabolism (Frederick *et al.*, 2000). Approximately 30% of the cases of autosomal dominant retinitis pigmentosa are now believed to be caused by nearly 100 different mutations in the rhodopsin gene (Gal *et al.*, 1997). The degeneration of rod photoreceptor cells could be caused by several mechanisms such as the inability of the mutant rhodopsin to fold correctly or to be transported properly in the cell or to activate transducin. Two rhodopsin mutants, P23H and G118R, were expressed in cultured COS cells. The first gave mixtures of the correctly folded and misfolded opsins while the second gave totally misfolded opsins (Liu *et al.*, 1996b; Garriga *et al.*, 1996), which presumably led to less compact disk membranes in the rod outer segment with the consequent degeneration of the photoreceptor cells. Mutation of Lys296 with Glu or Met did not affect a correct protein folding, but prevented the opsin from binding retinal. Both mutants gave constitutively active opsin, which however, was found phosphorylated and blocked by arrestin (Rim and Oprian, 1995; Li *et al.*, 1995).

Several mutations that affect the function of guanylate cyclase and the activator protein have been linked to various forms of congenital human retinal diseases, such as Leber congenital amaurosis, cone and rod-dystrophy, a disease that causes degeneration of cones followed by the death of rods (see the review article of Dizhoor, 2000).

Mutations in RPE65 have been discovered in autosomal recessive retinal dystrophy characterized by blindness at birth (Gu *et al.*, 1997). Moreover, the phenotype of RP65 knockout mice shows that RPE65 is directly involved in all-*trans*-to-11-*cis* isomerization, because RPE65-deficient retinal pigment epithelium does not produce 11-*cis* retinoids (Redmond *et al.*, 1998).

The human retinal disease of congenital night blindness does not lead to retinal degeneration, but inhibits the patient in adapting to darkness and seeing under dim light. Two mutations in rhodopsin have been shown to cause the disease, Ala292Glu and Gly90Asp (Dryja *et al.*, 1993; Sieving *et al.*, 1995). The two mutants, when expressed in transfected COS cells, gave opsins which bound the 11-*cis* retinal chromophore and were able to activate transducin in the light essentially as the wild-type rhodopsin. In the absence of retinal however, the mutated opsins activated transducin constitutively (Rao *et al.*, 1994). The cause of disease in patients having such mutations in rhodopsin, could be attributed to the fact that the photoreceptor cells adapt to the apparent internal light signal generated by mutant rhodopsin which lose retinal for thermal dissociation, and no longer responds to external stimuli (Rao and Oprian, 1996). The G38D mutation in the alpha chain of transducin is responsible for autosomal dominant night blindness (Dryja *et al.*, 1996). Residue 38 forms a critical part of the active site for GTP hydrolysis and diminishes GTPase activity, leading to excessive signaling in response to light, thereby elevating the background noise. Recessive loss-of-function mutations in either arrestin or rhodopsin kinase produce Oguchi disease, a variant of congenital stationary night blindness (Fuchs *et al.*, 1995; Cideciyan *et al.*, 1998).

Peripherin/RDS and rom-1 are subunits of a photoreceptor-specific integral disk membrane protein complex that is believed to maintain the

unique rim region of rod and cone outer segments (Molday, 1998). Many different heterozygous mutations in peripherin/RDS have been associated with human autosomal dominant retinal degenerations. In addition to the diversity of diseases described for humans, a defect in the RDS gene that prevents expression of peripherin/RDS is the cause of the retinal degeneration slow (RDS) phenotype in mice (Connell *et al.*, 1991). Recently, a transgenic mouse model for peripherin/RDS-based human disease has also been reported to undergo retinal degeneration (Kedzierski *et al.*, 1997).

7. INVERTEBRATE PHOTOTRANSDUCTION

The phototransduction in invertebrates shares a great deal of similarity with that in vertebrates in the overall strategy, but differs in the underlying molecular events. In most of the invertebrate photoreceptor cells, rhodopsin is embedded in microvilli-arranged rhabdomeric membranes. The absorption of light by rhodopsin leads to the opening of ion channels in the plasma membrane through which an influx of Na^+ and Ca^{2+} induces a depolarization of the photoreceptor membrane (Bacigalupo and Lisman, 1983). Despite the opposite results of the photon absorption on the ion channels, opening versus closing, invertebrate and vertebrate phototransduction cascades share several important features such as sensitivity over a large range of light intensity, high temporal resolution and a signal amplification of the order of 10^5 . However, no complete picture of phototransduction in invertebrate photoreceptors has yet emerged (see review articles: O'Day *et al.*, 1997; Lott *et al.*, 1999).

In contrast to vertebrates, invertebrate phototransduction utilizes the cascade of inositoltriphosphate (IP_3), which acts as a second messenger that releases calcium from intracellular stores (Berridge and Irvine, 1989). Upon absorption of a photon, the rhodopsin chromophore is isomerized from the 11-*cis* to the all-*trans* configuration and this leads to a change in the conformation of the protein which is transformed to metarhodopsin. As in vertebrates, metarhodopsin is the activated state of

rhodopsin and is able to activate a G-protein, which in turn activates a phospholipase C (PLC). PLC catalyzes the conversion of membrane phospholipids, phosphatidylinositol-4,5-bisphosphate (PIP₂) to inositol-1,3,5-triphosphate (IP₃) and diacylglycerol (DAG). IP₃ causes a rapid release of intracellular calcium ions from the submicrovillar cisternae, the organelle where calcium is sequestered (Payne and Fein, 1987), while DAG diffuses through the membrane bilayer where it activates protein kinase C. The local elevation of Ca²⁺ concentration has multiple physiological consequences because many molecular events in phototransduction are Ca²⁺-dependent. However, the precise mechanism by which stimulation of PLC and Ca²⁺ release from internal calcium stores leads to the opening of the light-sensitive channels, remain unresolved.

Invertebrate phototransduction has been studied mainly on the rhabdomeral photoreceptors of the horseshoe crab, *Limulus*, of the fruitfly *Drosophila* and of cephalopods such as *Loligo*.

Injection of IP₃ into *Limulus* ventral photoreceptors induces Ca²⁺ release (Brown and Rubin, 1984) and channel opening, unless intracellular calcium is buffered (Payne *et al.*, 1986), suggesting that Ca²⁺ is the intracellular messenger responsible for plasma membrane channel gating. However, as for vertebrates, there is a body of evidence that cGMP can act as a ligand to open light-sensitive channels in *Limulus* photoreceptors (Johnson *et al.*, 1986; Bacigalupo *et al.*, 1991; Dorlochter and de Vente, 2000). It has been suggested that IP₃ and cGMP in *Limulus* photoreceptors may work *via* independent, parallel pathways (Nagy, 1993; Rayer *et al.*, 1990), but the findings that the injection of calcium buffer (Shin *et al.*, 1993) or Ca²⁺/calmodulin-binding peptides (Richard *et al.*, 1997) totally blocked the *Limulus* photoreceptor light response, suggest instead that cGMP may function as a part of a single, sequential excitation pathway dependent on IP₃-mediated calcium release.

A powerful combination of electrophysiology and genetics has provided strong evidence that IP₃-mediated Ca²⁺ release is essential for phototransduction in microvillar photoreceptors of *Drosophila*. Mutations in the *norpA* gene which encode a phospholipase C render flies completely

blind (Bloomquist *et al.*, 1988) and photoreceptors of *norpA* mutant flies lack the light-induced inward Ca²⁺ current (Peretz *et al.*, 1994).

IP₃-mediated Ca²⁺ release from the calcium stores may directly activate the "transient reception potential" (TRP)-type of cation channels which are responsible for the light-induced Na⁺ and Ca²⁺ influx (Niemeyer *et al.*, 1996). However, the mechanism by which Ca²⁺ release is coupled to Na⁺ and Ca²⁺ entry remains elusive (see review articles: (Zucker, 1996; Montell, 1999).

Cyclic GMP may also participate in *Drosophila* phototransduction since cGMP-gated ion channels (Bauman *et al.*, 1994) and a soluble guanylate cyclase are expressed in the *Drosophila* compound eye (Shah and Hyde, 1995). In addition, the application of cGMP to *Drosophila* photoreceptor cells enhances the response to light (Bacigalupo *et al.*, 1995). The concentration of cGMP could be regulated by the intracellular IP₃-mediated Ca²⁺ increase which could modulate the activities of either cyclase and/or phosphodiesterase, as for the vertebrates. However, the physiological role of cGMP-gated channels remains to be established.

Also, the phototransduction in cephalopods is critically dependent on PLC activation. Rhodopsin is coupled to PLC by an α -type G protein subunit in squid and octopus retina (Kikkawa *et al.*, 1996). Moreover, in the squid photoreceptors the concentration of IP₃ increases rapidly upon illumination (Wood *et al.*, 1989). A TRP homologue has been isolated in squid (Monk *et al.*, 1996), but no cGMP-gated channels have so far been found in cephalopods eyes.

8. FUTURE DIRECTIONS

The molecular mechanism of phototransduction in the vertebrate eye is known in considerable detail. The participating proteins, present in ROS at high concentrations, have been well characterized at a molecular level. Moreover, the photoreceptor's electrical response to illumination has been studied extensively. At present, a quantitative and accurate description of the rising phase of the electrical signal in molecular terms based on stochastic simulations has been obtained (Lamb, 1996) while a comprehensive description of the

shutoff reactions for a complete account of the photoreceptor's response to light has still to be achieved. Therefore, many important details remain to be explained. For instance, the knowledge of the lifetime of Rh* is necessary for determining the recovery of the photoresponse. Rate-limiting values of 1–2 s for Rh* shut-off were found in tiger salamander rods (Pepperberg *et al.*, 1992), whereas other results give values much shorter and assign to G-protein deactivation the rate-limiting for shut-off of the phototransduction cascade (Sagoo and Lagnado, 1997). The decrease in Ca²⁺, via the action of recoverin, may increase the rate of Rh* phosphorylation effectively by decreasing the lifetime of Rh*. However, the roles of recoverin and Ca²⁺-dependent regulation of Rh* phosphorylation have not yet been resolved. Some experiments gave evidence that recoverin can mediate Ca²⁺-dependent modulation of rhodopsin kinase (Chen *et al.*, 1995) but more recent experiments gave the opposite result that rhodopsin phosphorylation is insensitive to changes in Ca²⁺ (Otto-Bruc *et al.*, 1998).

Another important goal for future research is the identification of amino acid changes that are responsible for the red shift of the retinal absorbance of different rhodopsins. A significant improvement will be observed by an increase in the number of amino acid sequences of a variety of opsins. Mutagenesis experiments must be carried out by considering the actual opsin sequences, so that the prediction of the effects of amino acid changes on the maximal absorbance shifts can be more accurate. This approach will improve the understanding of the molecular bases of the spectral tuning of visual pigments as well as the evolutionary processes taken by different species in adapting to their photic environment (see also the excellent review of Yokoyama, 2000).

The great challenge for the future will be the finding of an efficacious treatment of human retinal diseases, as at the moment there is no accepted treatment for the vast majority of monogenic retinopathies. Genetic approaches will identify biochemical pathways that may be relevant to the pathogenesis of diseases and the consequent therapy. The ultimate identification of all genes expressed in the retina will improve our knowledge of retinal diseases and will provide

the development of new strategies for their efficacious treatments.

The molecular events of phototransduction are known by their general outlines but only for vertebrates. The research groups in the vision field still have the possibility of exciting investigations to solve the puzzle of invertebrate phototransduction. The candidates for the internal transmitter so far are two: IP₃ and cGMP. Are they correlated in their action? What is the connection between IP₃-mediated Ca²⁺ release from the calcium stores and the depolarizing Na⁺ and Ca²⁺ influx? Calcium stores are located in the cell bodies at about 1.5 µm of distance from the rhabdomeres where the phototransduction proteins work. The diffusion of IP₃ and of a second signal back to the rhabdomeres after Ca²⁺ release, is difficult to reconcile with the millisecond activation of the light response. The problem reminds us of the strong dispute about the role of Ca²⁺ versus cGMP in vertebrate phototransduction in the seventies.

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